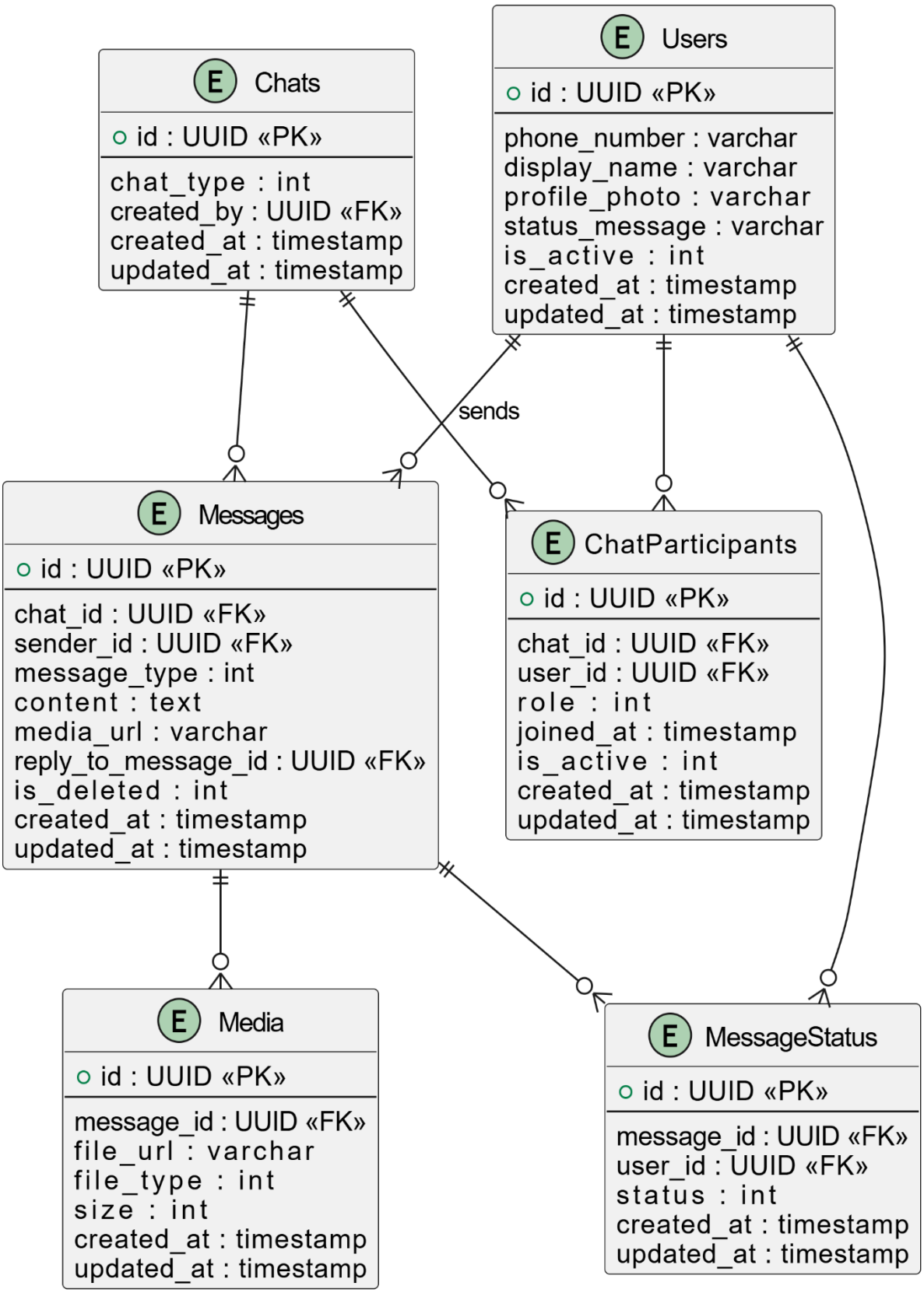
# ASSIGNMENT 03: WHATSAPP CHAT SYSTEM ERD



**System Overview**

This system represents the simple database design of a whatsApp - like messaging application. It involves tables like users, chats, messages, media, chat-participants, and message delivery status. Each table stores specific information to keep the database organized.

All records use UUID as primary keys, and integer values (0,1,2) are used for types and statuses instead of strings. This introduces optimization at database level.

## Chat Type

Defines the type of chat.

| Value | Meaning |
|---|---|
| 0 | private chat (1-to-1 conversation) |
| 1 | group chat |
| 2 | broadcast / Community message |

## Participant Role

Defines the role of a user inside a chat.

| Value | Meaning |
|---|---|
| 0 | member |
| 1 | admin |

## Message Type

Defines the type of message sent in a chat.

| Value | Meaning |
|---|---|
| 0 | text message |
| 1 | image |
| 2 | video |
| 3 | document |
| 4 | audio |

## Message Status

Represents the delivery status of a message.

| Value | Meaning |
|---|---|
| 0 | sent |
| 1 | delivered |
| 2 | read |

## Boolean Fields

Some fields use integers to represent true or false.

| Value | Meaning |
|---|---|
| 0 | false / inactive |
| 1 | true / active |

Example fields: **is_active, is_deleted**